INFORMATION SYSTEMS DEVELOPMENT

2022H1 OSE (Kasper, Shokoofeh)

Learning Objectives

- Getting familiar with the overall process of developing a new software application;

- Getting familiar with two software development methodologies;
  - SLDC (Systems-Development Life Cycle)
  - Agile

Information systems development

- The process of turning dreams into the reality.
  - The idea of adding a new functionality to an existing software system
  - The idea of designing a new business process which may need new hardware or software to support it!

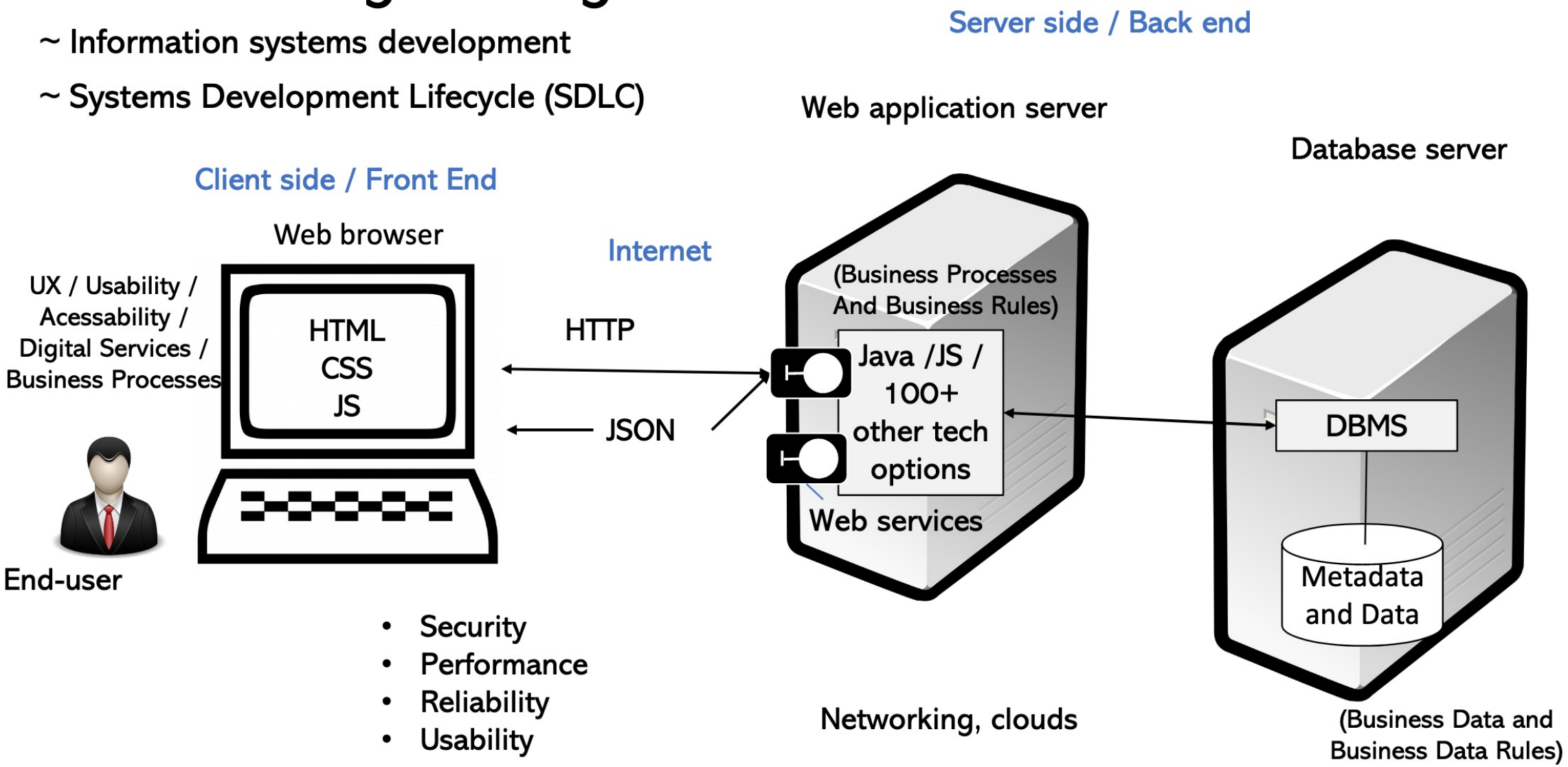
Software Engineering != Programming

- Programming => Coding
- A software application => “a **set** of instructions or programs that tell a computer what to do” [1].
- Software engineering => ” a **process** of analysing user requirements and then **designing**, **building**, and **testing** software application which will **satisfy** those requirements.”[2] .

Software engineering

~ Information systems development

~ Systems Development Lifecycle (SDLC)



- Security
- Performance
- Reliability
- Usability
- . . .

©2018 Juhani Välimäki

Specialties in Information systems development

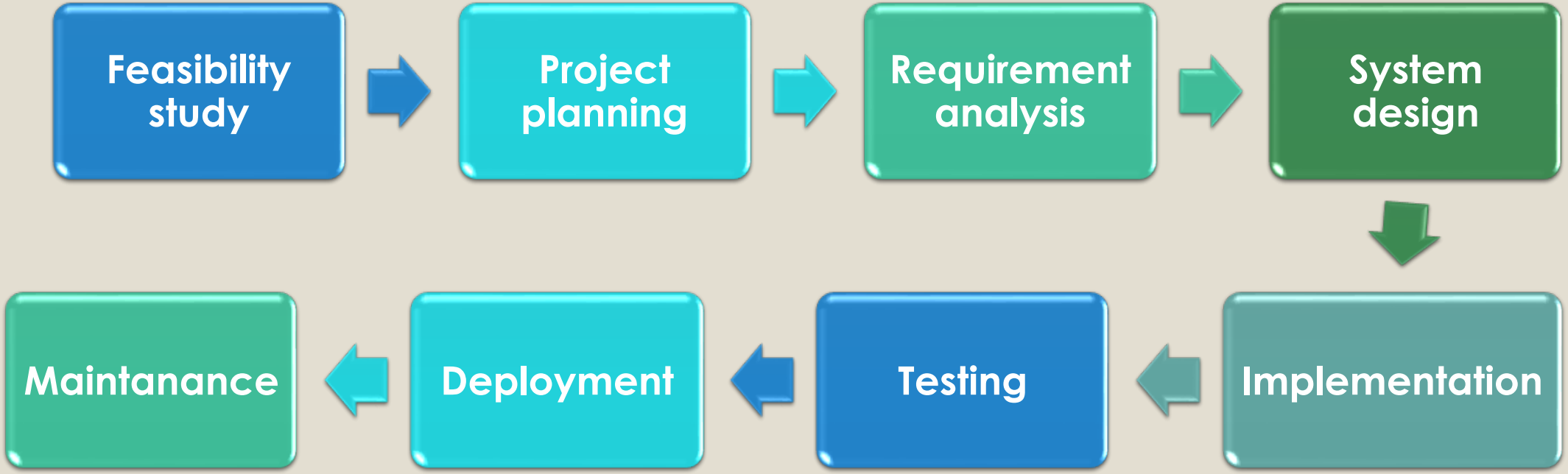
Computer programmer
Web developer
Systems analyst
Project manager
User experience designer
Database developer
Quality assurance tester
Network engineer
IT security specialist
AI developer
Data Analyst
Software architecture
And a lot more …[3]

SLDC (Systems-Development Life Cycle) [1]

- First developed in the 1960s
- Designed to manage large projects
- Replacing the old system with a new system

* The content of the following slides are mainly from the book "**Information Systems for Business and Beyond**", David T. Bourgeois, Ph.D. and Bourgeois [1]

SLDC (Systems-Development Life Cycle)



SLDC - Feasibility study

- Request review
  - Possibility of creating a solution
    - technical feasibility
    - the economic feasibility
    - legal feasibility
  - Existing alternatives

- **Result**: a decision on whether the project should even get started.

SLDC - Project planning

- Time frame
- Structuring teams
- Estimating the required budget

- **Result**: a plan for performing the project!

SLDC - Requirement analysis (System Analysis)

- Analysing the requirements of stakeholders
  - Interviews with the key players
  - Documenting the procedures
  - Deficiencies in the existing system
  - Needed features, functions and capabilities

- **Result**: system-requirements document!

SLDC – system design

- Translating **business requirements** into **specific technical requirements:**
  - user interface
  - database
  - data inputs and outputs
  - reports

- **Result**: system-design document!

SLDC – Implementation (Programming)

- Develop the program
  - Structuring the team of programmers

- **Result**: an initial working program that meets the requirements!

SLDC – Testing

- Test the developed the program
  - unit test
    - Testing every individual parts of the code for errors or bugs
  - system test
    - check if the components work together properly
  - user-acceptance test
    - Test by real system's users to ensure that it meets their standards.

- **Result**: A bug-free software application ready to use in the organization.

SLDC – Deployment

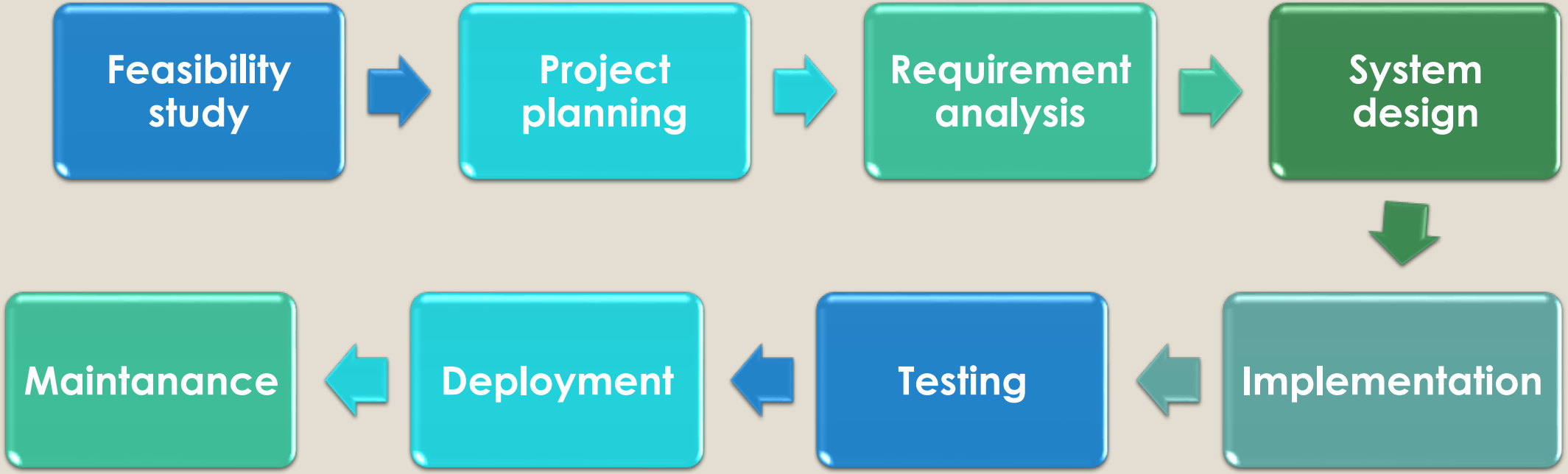
- Deploy the system in the organization
  - training the users
  - providing documentation
  - transfering from old system to the new one

- **Result**: The new software application is in use within the organization.

SLDC – Maintenance

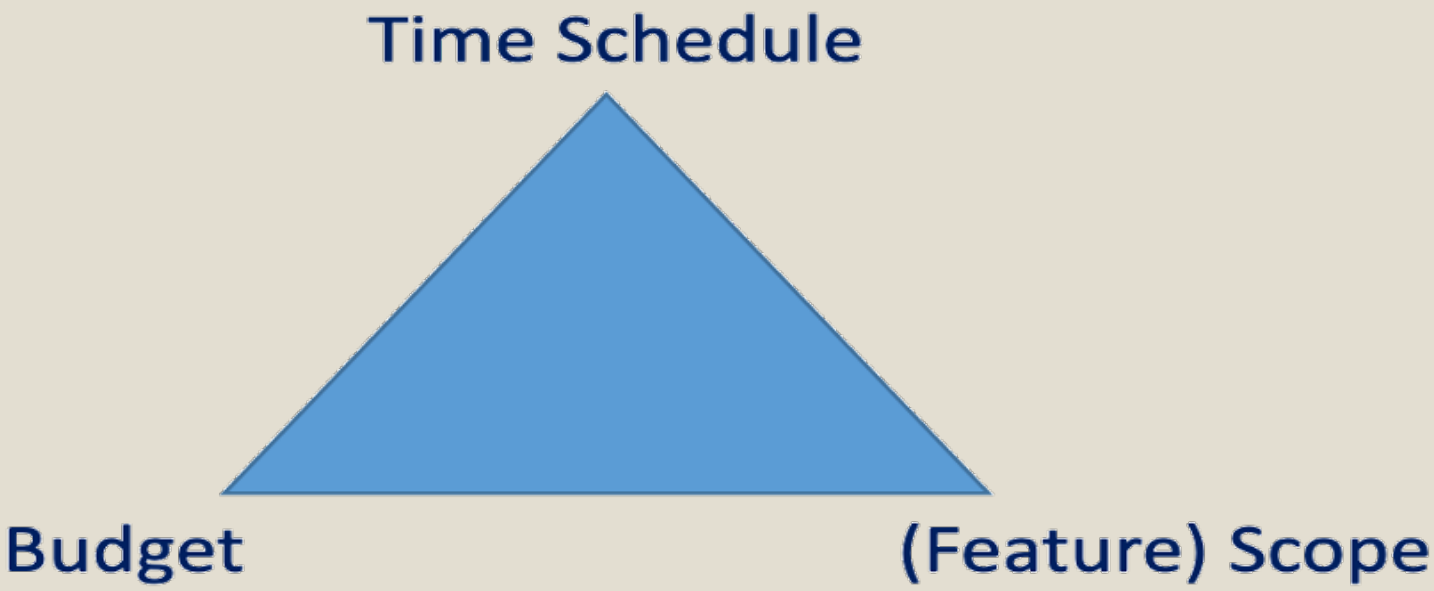
- Providing a structured support system
  - To fix the reported bugs
  - To evaluate the new requests and implement them
- System updates
- Backups

SLDC (Systems-Development Life Cycle)



SLDC – Constraints

- Need to fix the project planning triangle



References

[1] https://bus206.pressbooks.com/chapter/chapter-10-information-systems-development/

[2] https://www.guru99.com/what-is-software-engineering.html

[3] https://ca.indeed.com/career-advice/finding-a-job/careers-in-it

[4]https://www.techtarget.com/searchsoftwarequality/definition/systems-development-life-cycle